

# RUN-006 - VPN Access Provisioning (GlobalProtect)

## Purpose

Step-by-step procedure for provisioning, modifying, and revoking user access to the Summit IT Palo Alto GlobalProtect VPN. Covers standard employee access, elevated split-tunnel profiles, and troubleshooting common connectivity issues.

## Prerequisites / Access Needed

| Requirement                | Details                                                                     |
|----------------------------|-----------------------------------------------------------------------------|
| Approved REQ ticket        | Manager-authorized request for VPN access                                   |
| Okta Admin Console         | Role: User Admin - to add/remove Okta group membership                      |
| Palo Alto Panorama console | Read access for monitoring; write access for L2+ for gateway config changes |
| SummitDesk admin           | To log and close the ticket                                                 |

## VPN Architecture Overview

| Component       | Detail                                                                                             |
|-----------------|----------------------------------------------------------------------------------------------------|
| Client          | Palo Alto GlobalProtect agent                                                                      |
| Gateway         | <code>vpn.summitit.com</code> (primary);<br><code>vpn-west.summitit.com</code> (secondary)         |
| Portal          | <code>vpn.summitit.com</code>                                                                      |
| Auth method     | Okta SAML SSO + Okta Verify MFA (push)                                                             |
| Access profiles | Standard (full-tunnel, all corp traffic),<br>Split-Tunnel (specific routes only)                   |
| Managed via     | Palo Alto Panorama at<br><a href="https://panorama.summitit.com">https://panorama.summitit.com</a> |

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## Access Profiles Reference

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| Okta Group                    | VPN Profile          | Access Level                                                      |
|-------------------------------|----------------------|-------------------------------------------------------------------|
| VPN-GlobalProtect-Standard    | Standard Full-Tunnel | All corporate resources, internet via corp proxy                  |
| VPN-GlobalProtect-SplitTunnel | Split-Tunnel         | Corporate subnets only; internet direct                           |
| VPN-GlobalProtect-Vendor      | Vendor Limited       | Specific application subnets only; requires manager approval + L2 |
| VPN-GlobalProtect-Admin       | Admin Full-Tunnel    | Privileged access - L3 only; requires CISO approval               |

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## Procedure - Provisioning Standard VPN Access

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### 1. Validate the Request

1. Open the REQ ticket.
2. Confirm manager authorization is present in the ticket.
3. Verify the user is a current employee or authorized contractor in Okta (status: Active).
4. Confirm the access profile requested (Standard is default for all employees).

### 2. Add User to Okta VPN Group

VPN access is controlled entirely via Okta group membership. The GlobalProtect agent uses Okta SAML for authentication.

1. Log into <https://summitit.okta.com/admin>.
2. Go to **Directory > Groups**.
3. Search for `VPN-GlobalProtect-Standard`.
4. Click the group > **Manage People > + Add Member**.
5. Search for the user by UPN and add them.
6. Click **Save**.

Okta pushes the group change to the GlobalProtect policy within **5 minutes**.

For Split-Tunnel or Vendor profiles, add the user to the appropriate group instead. Vendor profile requires a second L2 approval comment in the ticket before adding.

### 3. Verify Okta Group Assignment

1. Go to **Directory > People > [User] > Groups** tab.
2. Confirm **VPN-GlobalProtect-Standard** (or the requested profile) appears in the group list.

#### 4. Confirm GlobalProtect Client is Installed

**Windows:** - GlobalProtect is deployed via Intune. Verify the agent is installed: check **C:\Program Files\Palo Alto Networks\GlobalProtect\** or search for "GlobalProtect" in the Start Menu. - If not installed: Intune > Devices > [Device] > **Apps** tab - find GlobalProtect and click **Install**.

**Mac:** - GlobalProtect is deployed via Jamf. Verify via Jamf Pro > Computers > [Device] > **Software** tab. - If not installed: Jamf Pro > Computers > [Device] > **Remote Commands > Install GlobalProtect**. - User can also install via **Jamf Self Service** by searching for "GlobalProtect".

#### 5. Configure GlobalProtect Client (First-Time Setup for User)

Advise the user to follow these steps once the agent is installed:

1. Open the GlobalProtect application (system tray / menu bar icon).
2. If prompted for a portal address, enter: **vpn.summitit.com**.
3. Click **Connect**.
4. A browser window opens - authenticate with Okta credentials (UPN + password).
5. Approve the Okta Verify push on their mobile device.
6. GlobalProtect connects automatically after MFA approval.
7. The system tray icon turns green with a shield - connection is active.

#### 6. Notify the User

Reply to the REQ ticket:

Subject: VPN Access Enabled - GlobalProtect

Hi [User Name],

Your VPN access has been provisioned.

To connect:

1. Open GlobalProtect (system tray or menu bar).
2. Portal address: **vpn.summitit.com**
3. Sign in with your Summit IT credentials (Okta).
4. Approve the Okta Verify push.

VPN is required when accessing internal resources (e.g., file shares, internal apps) from outside the office. On-network (Summit-Corp Wi-Fi), VPN is not required.

If you experience connection issues, see the IT self-service portal or contact:  
+1 (800) 555-7866 | servicedesk@summitit.com | Slack: #it-help

Ticket: [REQ-#####]  
Summit IT Service Desk

#### 7. Close the Ticket

1. Document: user UPN, Okta group assigned, date/time, your name.
2. Set status to **Resolved**.

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## Procedure - Revoking VPN Access

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Used during offboarding (also covered in RUN-003) or when access is no longer needed.

1. Log into <https://summitit.okta.com/admin>.
2. Go to **Directory > Groups > VPN-GlobalProtect-Standard** (or applicable group).
3. Find the user in the member list and click **Remove** (X icon).
4. Save.
5. In Panorama, verify the user has no active sessions: **Monitor > GlobalProtect > Connected Gateways** - search for the user.
6. If an active session is found, select it and click **Disconnect**.
7. Log the action in the ticket.

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## Troubleshooting Common VPN Issues

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### Issue: "Cannot connect to portal vpn.summitit.com"

| Check                 | Steps                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|
| Internet connectivity | Confirm user has internet access (browse to google.com)                                                                 |
| Portal address typo   | Re-enter exactly: <code>vpn.summitit.com</code>                                                                         |
| GlobalProtect version | Verify the installed version matches the current approved version (see Intune/Jamf app version)<br>- update if outdated |
| DNS resolution        | From command prompt: <code>nslookup vpn.summitit.com</code> - should resolve to a Palo Alto gateway IP                  |

### Issue: "Authentication failed" or redirect loop at Okta

| Check               | Steps                                                        |
|---------------------|--------------------------------------------------------------|
| Okta account status | Confirm user's Okta account is Active (not suspended/locked) |
| Okta VPN group      | Confirm user is in<br>VPN-GlobalProtect-Standard             |

|                      |                                                                            |
|----------------------|----------------------------------------------------------------------------|
| Okta Verify enrolled | Confirm MFA factor is enrolled; if not, re-enroll per RUN-001 Step 6       |
| Okta app assignment  | Okta Admin > Applications > GlobalProtect > check user appears as Assigned |

### Issue: Connected but cannot reach internal resources

| Check             | Steps                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| VPN profile       | Confirm user is assigned to the correct profile (Standard vs. Split-Tunnel)                                    |
| Specific resource | Check if the resource IP/subnet is included in the user's VPN profile routes                                   |
| CrowdStrike block | CrowdStrike Firewall may be blocking - check Falcon console > Firewall Management for any blocks on the device |
| Escalate to L3    | If routes and profile are correct but access fails, escalate to L3 - Networking team                           |

### Issue: GlobalProtect disconnects frequently / high latency

| Check                     | Steps                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary gateway saturated | Check Panorama > Monitor > GlobalProtect > Gateway Utilization - if >80%, advise user to use <code>vpn-west.summitit.com</code> as alternative |
| ISP instability           | Run <code>ping vpn.summitit.com -n 20</code> and check for packet loss                                                                         |
| Tunnel MTU                | Advise user to disable and re-enable GlobalProtect; if recurring, escalate to L3 Networking                                                    |

## Verification

| Check               | Expected Result                                                  |
|---------------------|------------------------------------------------------------------|
| Okta group assigned | VPN-GlobalProtect-Standard (or appropriate group) shows for user |

|                                    |                                                  |
|------------------------------------|--------------------------------------------------|
| GlobalProtect status               | Connected - green shield icon                    |
| Can reach internal resource (test) | e.g., ping 10.10.1.1 (internal gateway) responds |
| Panorama - active sessions         | Session visible under user's UPN                 |

## Rollback / If It Fails

| Symptom                                      | Action                                                                                                      |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Okta group sync not pushing to GlobalProtect | Wait 5 minutes; if still not working, check Okta > Palo Alto Networks GlobalProtect app > Provisioning logs |
| GlobalProtect installer fails on Windows     | Re-push via Intune: Devices > [Device] > Apps > Install; reboot device                                      |
| GlobalProtect installer fails on Mac         | Re-push via Jamf remote command or have user run via Self Service                                           |

## Escalation

| Condition                                    | Escalate To                               | Contact                          |
|----------------------------------------------|-------------------------------------------|----------------------------------|
| Gateway configuration changes needed         | L3 - Networking                           | networking@summitit.com          |
| Split-Tunnel or Vendor profile provisioning  | L2 - must co-approve                      | Submit L2 co-approval in ticket  |
| Admin Full-Tunnel profile (privileged VPN)   | L3 + CISO approval                        | ciso@summitit.com                |
| VPN gateway outage (multiple users affected) | L3 - Networking; declare P1 if widespread | networking@summitit.com; RUN-007 |
| Palo Alto vendor support                     | Vendor Support                            | See vendor-support-contacts.md   |